DENSE NON-ARCHIMEDEAN ENTIRE CURVES ON KUMMER SURFACES

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Abstract. For every prime p, we construct a smooth projective Kummer K3 surface over $\mathbb{C}_p$ admitting a Zariski-dense analytic map from the entire analytic line. The construction gives counterexamples to Garcia-Fritz and Pasten's conjecture [GFP25, Conjecture 1.8], since these surfaces have Kodaira dimension zero.

1. Introduction

The following conjecture is stated in [GFP25, Conjecture 1.8].

Conjecture. *Let $X/\mathbb{C}_p$ be a smooth projective variety. If there is a p-adic analytic map*

$$f\colon \mathbb{A}^{1,\mathrm{an}}_{\mathbb{C}_p} \longrightarrow X^{\mathrm{an}}$$

with Zariski-dense image, then $\kappa(X) = -\infty$.

Throughout, analytic spaces and morphisms are considered in the rigid analytic sense, and an denotes rigid analytification.

Let K be an algebraically closed complete non-Archimedean extension of $\mathbb{Q}_p$, with absolute value $|\cdot|$ extending the p-adic absolute value. For $q \in K^\times$ with $0 < |q| < 1$, the Tate elliptic curve $E(q)$ has a surjective analytic uniformization

$$u\colon \mathbb{G}^{\mathrm{an}}_{\mathrm{m},K} \longrightarrow E(q)^{\mathrm{an}}, \qquad E(q)(K) \simeq K^\times/q^{\mathbb{Z}}. \tag{1.1}$$

The map u is a local analytic isomorphism with kernel $q^{\mathbb{Z}}$; analytically, $E(q)^{\mathrm{an}}$ is the quotient $\mathbb{G}^{\mathrm{an}}_{\mathrm{m},K}/q^{\mathbb{Z}}$. See more details in [Tate95] and [Sil94].

Theorem 1.1. *Suppose $q_1, q_2 \in K^\times$ satisfy the following conditions*

$$0 < |q_i| < 1, \qquad q_1^m = q_2^n \quad (m, n \in \mathbb{Z}) \quad \Longrightarrow \quad m = n = 0. \tag{1.2}$$

Let $E_i = E(q_i)$ be the corresponding Tate elliptic curves, let $A = E_1 \times E_2$, and let $X = \mathrm{Km}(A)$ be the Kummer surface of A. Then there exists a Zariski-dense analytic map

$$f\colon \mathbb{A}^{1,\mathrm{an}}_K \longrightarrow X^{\mathrm{an}}.$$

The surface X is a smooth projective K3 surface, and $\kappa(X) = 0$.

Over $\mathbb{C}_p$, the construction also disproves the K3 dense-image conjecture attributed to Cherry in [GFP25, Section 1.6].

Corollary 1.2. *For every prime p, Theorem 1.1 applies over $K = \mathbb{C}_p$ with*

$$q_1 = p, \qquad q_2 = p(1+p).$$

Consequently the conjecture above is false for smooth projective varieties over $\mathbb{C}_p$, already for Kummer K3 surfaces.

Proof. We have $|q_1| = |q_2| = p^{-1}$. If $p^m = p^n(1+p)^n$, then taking absolute values gives $p^{-m} = p^{-n}$, so $m = n$. It follows that $(1+p)^n = 1$, so $n = 0$ and therefore $n = m = 0$. Thus (1.2) holds, and Theorem 1.1 gives the result. □

For the rest of the construction we work under the assumption (1.2).

2. Tate uniformizations and the Kummer construction

We first recall the Kummer construction. The abelian surface $A = E_1 \times E_2$ has the involution

$$\iota = [-1]_A, \qquad (x_1, x_2) \longmapsto (-x_1, -x_2).$$

Its fixed locus is $A[2] = E_1[2] \times E_2[2]$. Since K has characteristic zero, $E_i[2](K) \simeq (\mathbb{Z}/2\mathbb{Z})^2$, so there are sixteen fixed points. The quotient $A/\langle\iota\rangle$ has an ordinary double point at the image of each of them. Its minimal resolution is the Kummer surface $\mathrm{Km}(A)$, a smooth projective K3 surface [Huy16, Chapter 1, Example 1.3(iii)]. Equivalently, blow up all sixteen points and then take the quotient by the lifted involution:

$$B = \mathrm{Bl}_{A[2]} A, \qquad \beta\colon B \longrightarrow A, \qquad X = \mathrm{Km}(A) = B/\langle\widetilde{\iota}\rangle.$$

We will use this description to construct an analytic map directly into the smooth surface X.

Let

$$u_i \colon \mathbb{G}_{\mathrm{m},K}^{\mathrm{an}} \longrightarrow E_i^{\mathrm{an}} \qquad (i = 1, 2)$$

be the uniformizations in (1.1), with $\ker u_i = q_i^{\mathbb{Z}}$. Define

$$h \colon \mathbb{G}_{\mathrm{m},K}^{\mathrm{an}} \longrightarrow A^{\mathrm{an}}, \qquad h(t) = \big(u_1(t), u_2(t)\big).$$

Lemma 2.1. *The image of h is Zariski dense in A.*

Proof. Let $Z \subseteq A$ be a Zariski-closed subset containing $h(K^\times)$. For each $a \in E_1(K)$, choose $t \in K^\times$ such that $u_1(t) = a$. Then, for every $n \in \mathbb{Z}$,

$$h(tq_1^n) = \big(u_1(tq_1^n), u_2(tq_1^n)\big) = \big(a, u_2(tq_1^n)\big) \in Z \cap \big(\{a\} \times E_2\big).$$

If $u_2(tq_1^n) = u_2(tq_1^{n'})$, then $q_1^{n-n'} = q_2^m$ for some $m \in \mathbb{Z}$. By (1.2), $n = n'$ and $m = 0$. Thus these points form an infinite subset of the irreducible curve $\{a\} \times E_2$. Every proper closed subset of that curve is finite, so Z contains the entire fiber $\{a\} \times E_2$. This holds for every $a \in E_1(K)$; hence Z contains $A(K)$. As K is algebraically closed, $A(K)$ is Zariski dense in A, and therefore $Z = A$. □

Let τ be the inversion on $\mathbb{G}_{\mathrm{m},K}^{\mathrm{an}}$, so $\tau(t) = t^{-1}$. We have $h \circ \tau = \iota^{\mathrm{an}} \circ h$ and the following square commutes:

$$\begin{array}{ccc} \mathbb{G}_{\mathrm{m},K}^{\mathrm{an}} & \xrightarrow{\ h\ } & A^{\mathrm{an}} \\ {\scriptstyle\tau}\downarrow & & \downarrow{\scriptstyle\iota^{\mathrm{an}}} \\ \mathbb{G}_{\mathrm{m},K}^{\mathrm{an}} & \xrightarrow[\ h\ ]{} & A^{\mathrm{an}}. \end{array}$$

Lemma 2.2. *The inverse image of $A[2]$ under h consists precisely of $t = \pm 1$. If $\mathcal{I}$ is the ideal sheaf of $A[2]$ in A, then its image ideal under pullback by h is*

$$\mathcal{I}\mathcal{O}_{\mathbb{G}_{\mathrm{m},K}^{\mathrm{an}}} = \mathcal{O}_{\mathbb{G}_{\mathrm{m},K}^{\mathrm{an}}}\big(-[1] - [-1]\big). \tag{2.1}$$

In particular, this image ideal is invertible.

Proof. Since $A[2] = E_1[2] \times E_2[2]$, for $t \in K^\times$ we have

$$\begin{aligned} h(t) \in A[2] \quad &\Longleftrightarrow \quad 2u_i(t) = 0 \quad (i = 1, 2) \\ &\Longleftrightarrow \quad u_i(t^2) = 0 \quad (i = 1, 2) \\ &\Longleftrightarrow \quad t^2 = q_1^m = q_2^n \quad \text{for some } m, n \in \mathbb{Z}. \end{aligned}$$

Condition (1.2) gives $m = n = 0$, hence $t^2 = 1$. Conversely, $t = \pm 1$ clearly maps to $A[2]$.

Since u_1 is a local analytic isomorphism, $u_1^{-1}(E_1[2])$ is a discrete union of rigid points. As K is algebraically closed, these rigid points are K-rational, so the calculation above determines the entire inverse image and there are no extra nonclassical points.

At $\varepsilon \in \{1, -1\}$, choose a local parameter x_1 on E_1 at $u_1(\varepsilon)$. Its pullback $u_1^* x_1$ has a simple zero at ε, because u_1 is a local analytic isomorphism. The pullback of x_1 belongs to the image ideal of the center, and every element of that ideal vanishes at ε. Consequently the image ideal in the local ring is generated by $t - \varepsilon$. Away from ± 1, it is the unit ideal. This proves (2.1). □

3. The equivariant lift

The involution ι preserves $A[2]$, so it lifts to an involution $\widetilde{\iota}$ on $B = \mathrm{Bl}_{A[2]} A$ satisfying $\beta \circ \widetilde{\iota} = \iota \circ \beta$. Let

$$\rho \colon B \longrightarrow X = B/\langle \widetilde{\iota} \rangle$$

be the finite surjective quotient map.

Lemma 3.1. *There is a unique analytic lift*

$$\widetilde{h} \colon \mathbb{G}_{\mathrm{m},K}^{\mathrm{an}} \longrightarrow B^{\mathrm{an}}, \qquad \beta^{\mathrm{an}} \circ \widetilde{h} = h.$$

It satisfies $\widetilde{h} \circ \tau = \widetilde{\iota}^{\mathrm{an}} \circ \widetilde{h}$ and has Zariski-dense image in B.

Recall that the blowup of a space along an ideal is universal for maps on which the image ideal becomes invertible. The same property holds for the analytification of an algebraic blowup, as one sees from the usual local blowup charts.

Proof. By Lemma 2.2, the image ideal of the center under h is invertible. The universal property therefore gives the unique lift:

$$\begin{array}{ccc} & & B^{\mathrm{an}} \\ & \overset{\exists!\,\widetilde{h}}{\nearrow} & \big\downarrow{\scriptstyle \beta^{\mathrm{an}}} \\ \mathbb{G}_{\mathrm{m},K}^{\mathrm{an}} & \xrightarrow[h]{} & A^{\mathrm{an}}. \end{array}$$

Furthermore,

$$\beta^{\mathrm{an}} \circ \widetilde{h} \circ \tau = h \circ \tau = \iota^{\mathrm{an}} \circ h = \beta^{\mathrm{an}} \circ \widetilde{\iota}^{\mathrm{an}} \circ \widetilde{h}.$$

Thus $\widetilde{h} \circ \tau$ and $\widetilde{\iota}^{\mathrm{an}} \circ \widetilde{h}$ are two lifts of the same map $\iota^{\mathrm{an}} \circ h$. Uniqueness gives their equality. The equivariance and lifting identities are summarized by

$$\begin{array}{ccccc} \mathbb{G}_{\mathrm{m},K}^{\mathrm{an}} & \xrightarrow{\widetilde{h}} & B^{\mathrm{an}} & \xrightarrow{\beta^{\mathrm{an}}} & A^{\mathrm{an}} \\ \big\downarrow{\scriptstyle \tau} & & \big\downarrow{\scriptstyle \widetilde{\iota}^{\mathrm{an}}} & & \big\downarrow{\scriptstyle \iota^{\mathrm{an}}} \\ \mathbb{G}_{\mathrm{m},K}^{\mathrm{an}} & \xrightarrow[\widetilde{h}]{} & B^{\mathrm{an}} & \xrightarrow[\beta^{\mathrm{an}}]{} & A^{\mathrm{an}}. \end{array}$$

Let $W \subseteq B$ be the algebraic Zariski closure of the image of $\widetilde{h}$. Since β is proper, $\beta(W)$ is closed in A. It contains the image of h, which is dense by Lemma 2.1; hence $\beta(W) = A$. It follows that $\dim W \geq \dim A = 2$. The irreducible surface B has no proper closed subset of dimension two, so $W = B$. □

We obtain an analytic morphism

$$g = \rho^{\mathrm{an}} \circ \widetilde{h} \colon \mathbb{G}_{\mathrm{m},K}^{\mathrm{an}} \longrightarrow X^{\mathrm{an}}. \tag{3.1}$$

It is invariant under τ, since $\rho \circ \widetilde{\iota} = \rho$; thus $g(t^{-1}) = g(t)$. It is also Zariski dense. Indeed, if an algebraic closed subset $D \subseteq X$ contains its image, then $\rho^{-1}(D)$ contains the dense image of $\widetilde{h}$, so $\rho^{-1}(D) = B$. Surjectivity of ρ implies $D = X$.

4. Descent to the entire affine line

Consider the algebraic morphism

$$\sigma \colon \mathbb{G}_{\mathrm{m},K} \longrightarrow \mathbb{A}^1_K, \qquad z = t + t^{-1}. \tag{4.1}$$

It is finite and flat of degree two, since

$$K[t, t^{-1}] \simeq K[z, T]/(T^2 - zT + 1)$$

is free over $K[z]$ with basis $1, T$. Inversion sends T to $T^{-1} = z - T$. If $a(z) + b(z)T$ is invariant, then

$$a(z) + b(z)T = a(z) + b(z)(z - T) = a(z) + b(z)z - b(z)T.$$

Comparing coefficients of T gives $2b(z) = 0$, and hence $b(z) = 0$. Therefore

$$K[t, t^{-1}]^{\langle \tau \rangle} = K[t + t^{-1}] = K[z], \qquad \mathbb{G}_{\mathrm{m},K}/\langle \tau \rangle \simeq \mathbb{A}^1_K.$$

The fixed points $t = \pm 1$ lie above $z = \pm 2$.

Lemma 4.1. *Let Y be a separated rigid analytic K-space. If $v \colon \mathbb{G}_{\mathrm{m},K}^{\mathrm{an}} \to Y$ is invariant under inversion, then there is a unique analytic morphism $\overline{v} \colon \mathbb{A}_K^{1,\mathrm{an}} \to Y$ such that $v = \overline{v} \circ \sigma^{\mathrm{an}}$:*

$$\begin{array}{ccc} \mathbb{G}_{\mathrm{m},K}^{\mathrm{an}} & \xrightarrow{v} & Y \\ \big\downarrow{\scriptstyle \sigma^{\mathrm{an}}} & \overset{\exists!\,\overline{v}}{\nearrow} & \\ \mathbb{A}_K^{1,\mathrm{an}}. & & \end{array}$$

The target Y need not arise from an algebraic variety, so the assertion uses the categorical quotient in the rigid analytic category.

Proof. By the finite-group quotient theorem and its compatibility with analytification [Zav24, Theorems 4.3.4 and 4.5.3],

$$\mathbb{G}_{\mathrm{m},K}^{\mathrm{an}}/\langle \tau \rangle \simeq (\mathbb{G}_{\mathrm{m},K}/\langle \tau \rangle)^{\mathrm{an}} \simeq \mathbb{A}_K^{1,\mathrm{an}}.$$

We apply these results through the fully faithful passage from rigid spaces to their associated adic spaces over $\mathrm{Spa}(K, K^\circ)$: the affine scheme $\mathbb{G}_{\mathrm{m},K}$ satisfies the algebraic orbit condition, and Theorem 4.5.3 then supplies the analytic orbit condition required by Theorem 4.3.4. Under this identification, the quotient map is σ^{an}. Its categorical universal property gives the bijection

$$\mathrm{Hom}_K(\mathbb{A}_K^{1,\mathrm{an}}, Y) \xrightarrow{\sim} \mathrm{Hom}_K(\mathbb{G}_{\mathrm{m},K}^{\mathrm{an}}, Y)^{\langle\tau\rangle}, \qquad \overline{v} \longmapsto \overline{v} \circ \sigma^{\mathrm{an}}.$$

This proves existence and uniqueness.

Let $R \in |K^\times|$, $R \geq 1$, and let $D_R = \{|z| \leq R\}$ be the closed disc of radius R. These discs exhaust $\mathbb{A}_K^{1,\mathrm{an}}$ as R tends to infinity. The inverse image $C_R = (\sigma^{\mathrm{an}})^{-1}(D_R)$ is

$$C_R = \{R^{-1} \leq |t| \leq R\}. \tag{4.2}$$

To verify this, there are three cases. If $|t| > 1$, then $|t^{-1}| < 1 < |t|$, so $|t + t^{-1}| = |t|$. If $|t| < 1$, then $|t + t^{-1}| = |t^{-1}| = |t|^{-1}$. If $|t| = 1$, then $|t + t^{-1}| \leq 1 \leq R$. In each case, $|z| \leq R$ is equivalent to the bounds in (4.2).

Let $A_R = \mathcal{O}(D_R) = K\langle z/R\rangle$ be the Tate algebra of this closed disc. Since C_R is the analytic base change of the finite map (4.1),

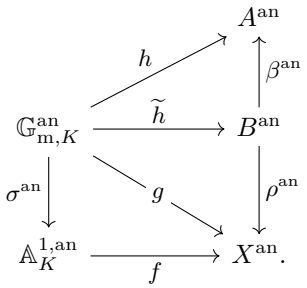
$$\mathcal{O}(C_R) = A_R[T]/(T^2 - zT + 1) \simeq A_R \oplus A_R T$$

and the τ-invariants are

$$\mathcal{O}(C_R)^{\langle\tau\rangle} = A_R = \mathcal{O}(D_R).$$

The categorical universal property above gives the unique descents to D_R. These agree on common smaller discs by uniqueness and recover the global map $\overline{v}$. □

Proof of Theorem 1.1. Apply Lemma 4.1 to the invariant morphism g of (3.1). We obtain the unique analytic map $f\colon \mathbb{A}_K^{1,\mathrm{an}} \to X^{\mathrm{an}}$ satisfying $g = f \circ \sigma^{\mathrm{an}}$, thus we have the following commutative diagram:



The image of g is contained in the image of f and is Zariski dense in X. Therefore f is Zariski dense. Since X is a smooth projective K3 surface $\kappa(X) = 0$. □

5. Remarks and consequences

Remark 5.1. The constructed map f cannot extend across infinity to an analytic map $(\mathbb{P}_K^1)^{\mathrm{an}} \to X^{\mathrm{an}}$. Indeed, proper GAGA would algebraize such a map. Its algebraic image would have dimension at most one, whereas it would contain the Zariski-dense image of f in the surface X, a contradiction.

Corollary 5.2. *There exist smooth projective varieties over $\mathbb{C}_p$ admitting Zariski-dense entire curves which are neither uniruled nor unirational.*

Proof. Use the Kummer surface X constructed in Corollary 1.2. Since unirationality implies uniruledness, it suffices to prove that X is not uniruled.

Suppose that X were uniruled, then there would be a generically finite dominant rational map

$$\varphi_0\colon C \times \mathbb{P}^1 \dashrightarrow X$$

for a smooth projective curve C. Resolving indeterminacy gives a smooth projective surface Y, a blow-up $\pi\colon Y \to C \times \mathbb{P}^1$, and a generically finite dominant morphism $\varphi\colon Y \to X$, and in particular generically separable, thus the pullback of differential forms is therefore injective:

$$\varphi^*\colon H^0(X, K_X) \hookrightarrow H^0(Y, K_Y).$$

As $K_X \simeq \mathcal{O}_X$ and X is geometrically connected, $H^0(X, K_X) \simeq K \neq 0$.

On the other hand,

$$\begin{aligned} H^0(Y, K_Y) &\simeq H^0(C \times \mathbb{P}^1, K_{C\times\mathbb{P}^1}) \\ &\simeq H^0(C, K_C) \otimes_K H^0(\mathbb{P}^1, K_{\mathbb{P}^1}) \end{aligned}$$

$$\simeq H^0(C, K_C) \otimes_K H^0(\mathbb{P}^1, \mathcal{O}_{\mathbb{P}^1}(-2)) = 0.$$

This is a contradiction. □

In particular these examples give a negative answer to the unirationality question in [GFP25, Question 1.9].

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